job_12-Evaluation report

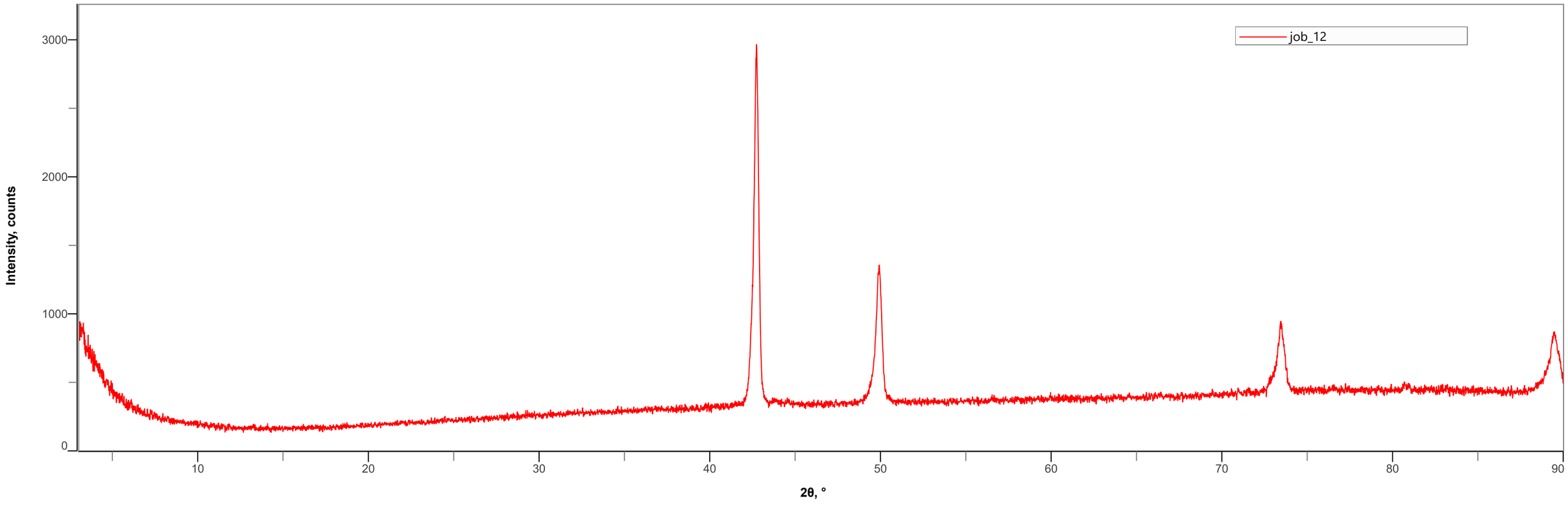
General information

Analysis date	2025-11-27 14:24:37	Measurement start time	2025-11-23 11:27:46
Analyst	Administrator	Operator	Administrator
Sample name	job_12	Comment	
Measured data name	C:\Users\DELL\Desktop\251\Vasanth\job_12.rasx	Memo	

Measurement Conditions

X-Ray generator	40 kV, 15 mA	Scan mode	1D(scan)
Incident primary	No unit	Scan speed/Duration time	10.00 °/min
Goniometer	MiniFlex 300/600	Step width	0.01 °
Attachment	ASC-8	Scan axis	θ/2θ
Filter	None	Scan range	3 ~ 90 °
Selection slit	None	DS	1.25deg
Diffracted beam mono	None	IHS	10 mm
Detector	HyPix-400 MF	SS	Open
Optics attribute	None	RS	Open

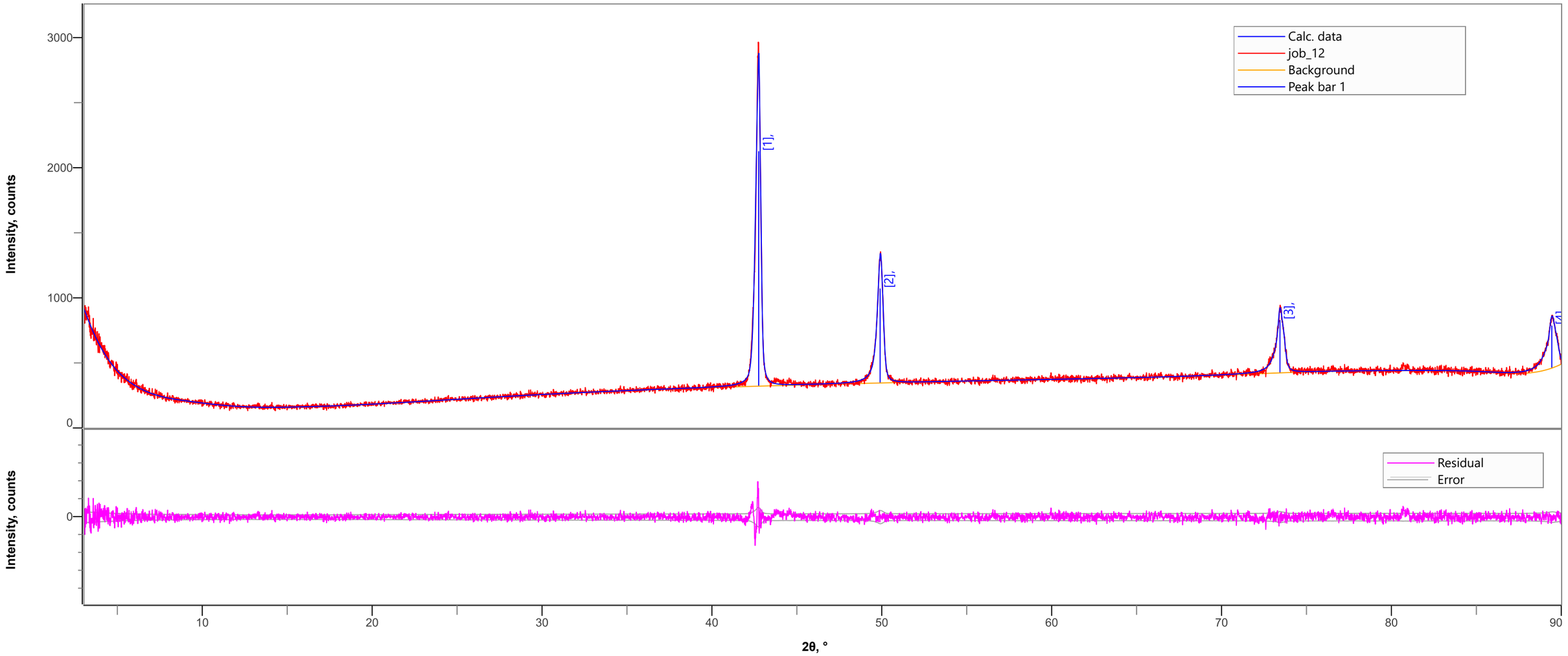
Measured profile view



Peak evaluation conditions

Peak search method	Second derivative method	σ cut	3.00		
Profile fitting	Run completed	Peak shape	Split pseudo-Voigt	Fitting condition	Auto(Refine background)

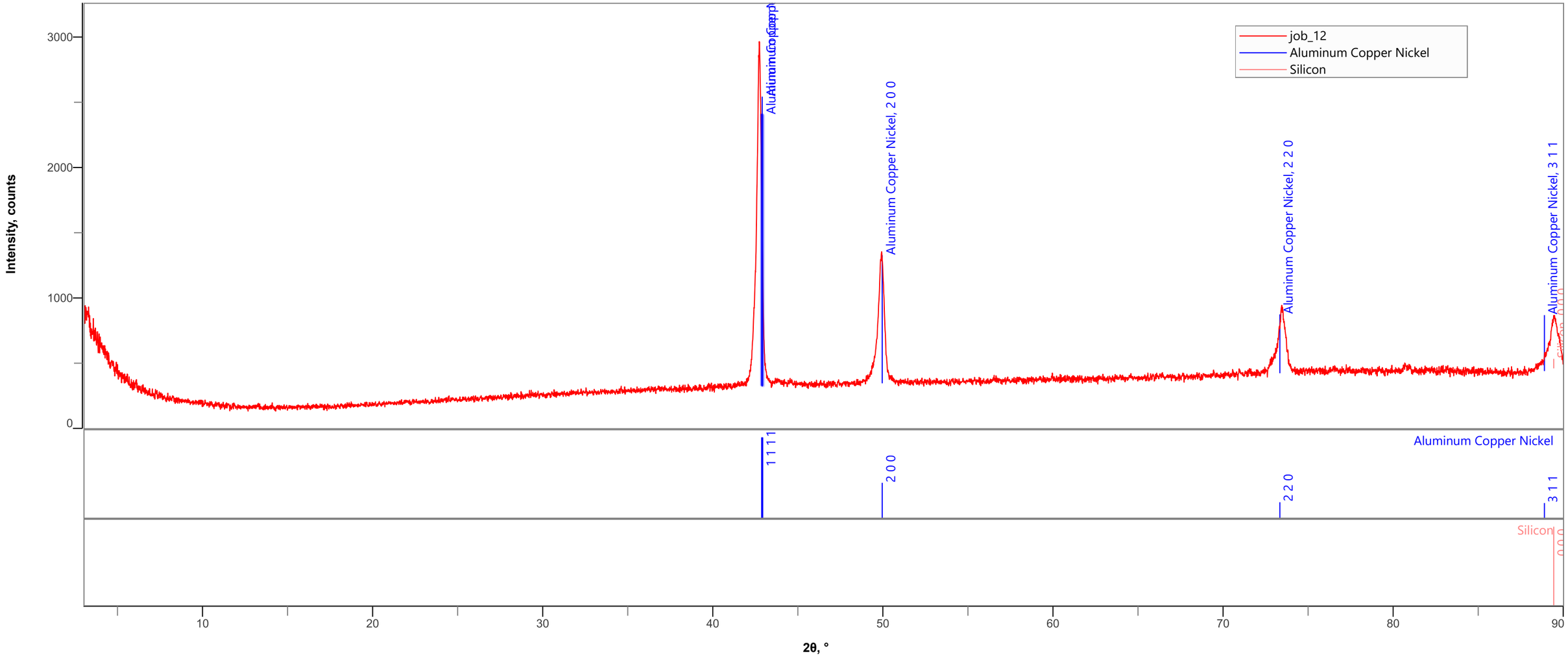
Peak Profile View



Qualitative Analysis Results

Phase name	Chemical formula	FOM	Phase reg. detail	Space Group	DB Card Number
Aluminum Copper Nickel	Cu0.03 Ni0.10 Al0.87	0.309	S/M:PDF-5+ 2026	225 : Fm-3m	04-008-0981
Silicon	Si	0.166	S/M:PDF-5+ 2026		00-047-1187

Phase Data View



Peak list

No.	2θ, °	d, Å	Height, counts	FWHM, °	Int. I., counts°	Int. W., °	Asymmetry	Decay (ηL/mL)	Decay(ηH/mH)	Size, Å	Phase Name	Chemical Formula	Card No	Norm. I.	Profile Type	Distribution R.S.D.	Strain, %	Degree of Orientation	Ring Factor	β Cluster	Custom Label
1	42.732(3)	2.11431(12)	1802(33)	0.311(2)	706(3)	0.392(9)	2.55(11)	0.364(13)	0.73(4)	286.1(19)	Aluminum Copper Nickel: 1 1 1	Cu0.03 Ni0.10 Al0.87	04-008-0981	100.00	Split Pseudo-Voigt	-	-	-	-	-	
2	49.876(6)	1.8269(2)	725(20)	0.346(5)	352(2)	0.485(17)	1.51(12)	0.86(4)	0.57(5)	265(4)	Aluminum Copper Nickel: 2 0 0	Cu0.03 Ni0.10 Al0.87	04-008-0981	49.78	Split Pseudo-Voigt	-	-	-	-	-	
3	73.397(7)	1.28897(10)	405(13)	0.343(12)	229(3)	0.56(3)	1.26(12)	1.35(6)	0.66(8)	302(11)	Aluminum Copper Nickel: 2 2 0	Cu0.03 Ni0.10 Al0.87	04-008-0981	32.40	Split Pseudo-Voigt	-	-	-	-	-	
4	89.391(8)	1.09520(8)	326(11)	0.427(15)	216(3)	0.66(3)	1.27(12)	1.39(6)	0.03(9)	273(10)	Silicon	Si	00-047-1187	30.60	Split Pseudo-Voigt	-	-	-	-	-	

Lattice parameters

Phase name	a, Å	b, Å	c, Å	α, °	β, °	γ, °
Aluminum Copper Nickel	3.65004	3.65004	3.65004	90.000	90.000	90.000
Silicon	0.00000	0.00000	0.00000	90.000	90.000	90.000

d-I List

Aluminum Copper Nickel

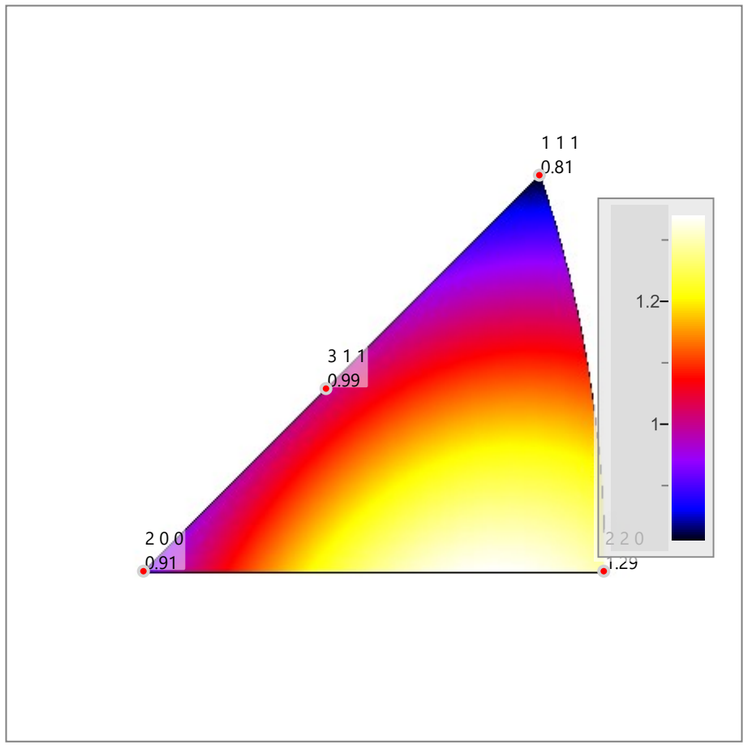
No.	2θ, °	d, Å	h k l	Norm. I.
1	42.88013	2.10735	1 1 1	100.00
2	49.93134	1.82502	2 0 0	46.48
3	73.29702	1.29049	2 2 0	25.84
4	88.84316	1.10053	3 1 1	29.20

Silicon

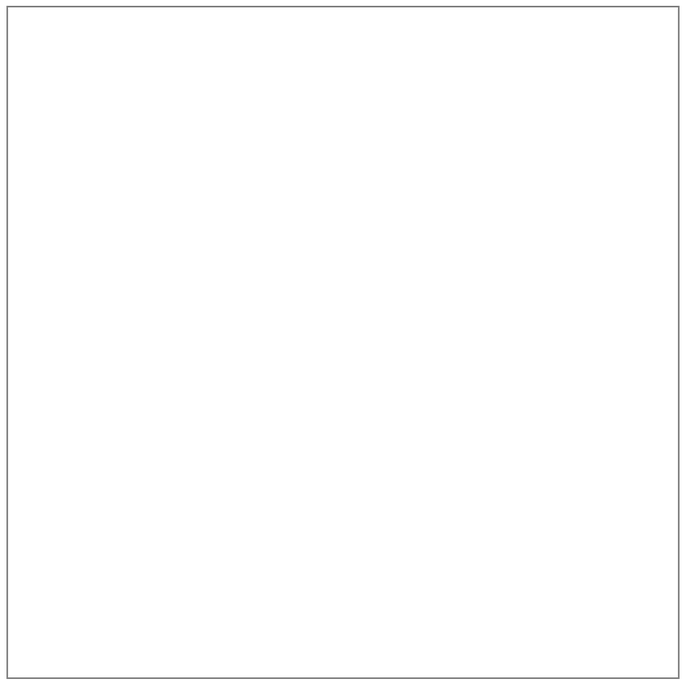
No.	2θ, °	d, Å	h k l	Norm. I.
1	89.61940	1.09300	0 0 0	100.00

Simple Inverse Pole Figure

Aluminum Copper Nickel

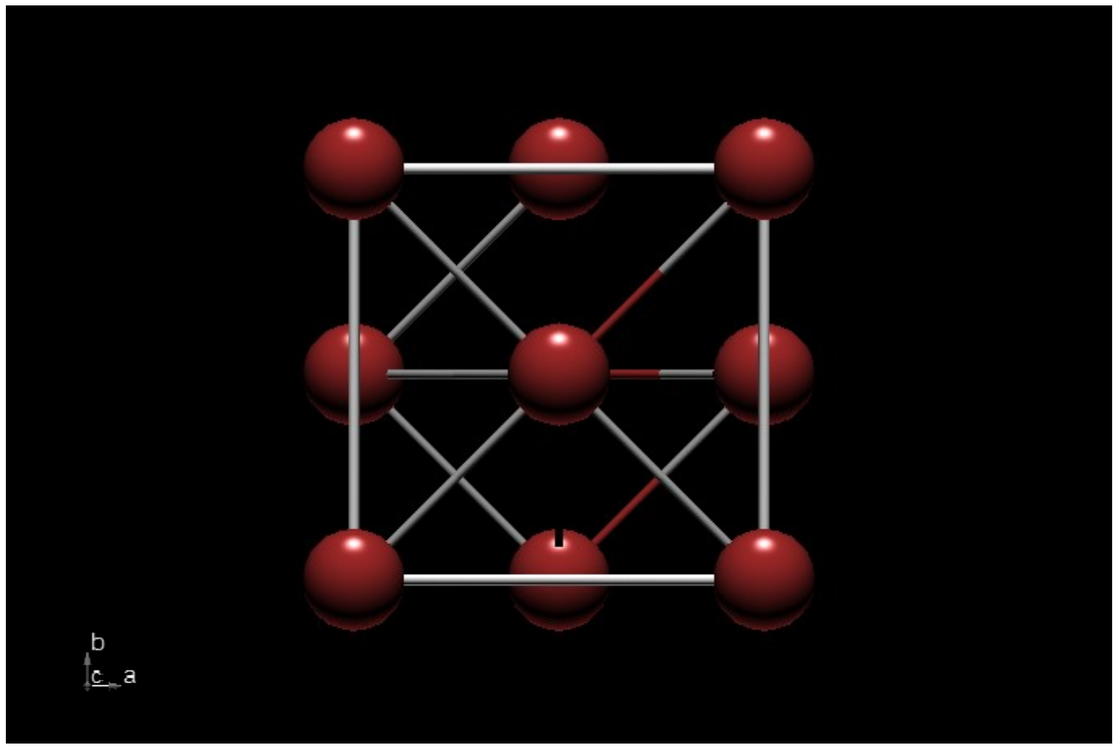


Silicon



Crystal Structure

Aluminum Copper Nickel



Structure parameters

Element	x	y	z	Occ.	B	Charge	Multiplicity	Valence	Bond Valence Sum
Cu1(Cu)	0.000000	0.000000	0.000000	0.030	0.500	0	4	0.000	0.000
Ni1(Ni)	0.000000	0.000000	0.000000	0.100	0.500	0	4	0.000	0.000
Al1(Al)	0.000000	0.000000	0.000000	0.870	0.500	0	4	0.000	0.000

Distance

Bond Def.	Distance, Å
Cu1-Al1(x,1/2+y,1/2+z)	2.58097
Cu1-Al1(x,-1/2+y,1/2+z)	2.58097
Cu1-Al1(x,1/2+y,-1/2+z)	2.58097
Cu1-Al1(1/2+x,1/2+y,z)	2.58097
Cu1-Al1(-1/2+x,1/2+y,z)	2.58097
Cu1-Al1(1/2+x,-1/2+y,z)	2.58097
Cu1-Al1(1/2+x,y,1/2+z)	2.58097
Cu1-Al1(-1/2+x,y,1/2+z)	2.58097
Cu1-Al1(1/2+x,y,-1/2+z)	2.58097
Ni1-Al1(x,1/2+y,1/2+z)	2.58097

Distance

Bond Def.	Distance, Å
Ni1-Al1(x,-1/2+y,1/2+z)	2.58097
Ni1-Al1(x,1/2+y,-1/2+z)	2.58097
Ni1-Al1(1/2+x,1/2+y,z)	2.58097
Ni1-Al1(-1/2+x,1/2+y,z)	2.58097
Ni1-Al1(1/2+x,-1/2+y,z)	2.58097
Ni1-Al1(1/2+x,y,1/2+z)	2.58097
Ni1-Al1(-1/2+x,y,1/2+z)	2.58097
Ni1-Al1(1/2+x,y,-1/2+z)	2.58097
Al1-Al1(x,1/2+y,1/2+z)	2.58097
Al1-Al1(x,-1/2+y,1/2+z)	2.58097
Al1-Al1(x,1/2+y,-1/2+z)	2.58097
Al1-Al1(1/2+x,1/2+y,z)	2.58097
Al1-Al1(-1/2+x,1/2+y,z)	2.58097
Al1-Al1(1/2+x,-1/2+y,z)	2.58097
Al1-Al1(1/2+x,y,1/2+z)	2.58097
Al1-Al1(-1/2+x,y,1/2+z)	2.58097
Al1-Al1(1/2+x,y,-1/2+z)	2.58097